Additional Exercise 2 · Exam type

T3 Additional Exercise 2 – INFOPYMES, statements 2016-2017 and ROA/ROE

Original statement · T3_Additional_EXERCISES.pdf, p. 3

Three entrepreneurs established INFOPYMES on 1/01/2016 to provide on-line information services for small and medium enterprises, start-ups and potential entrepreneurs.

- The computer equipment, worth 100,000 €, was acquired on 1/01/2016 and will be paid to suppliers after three months, on 1/04/2016. Useful life 5 years (until 31/12/2020), linear depreciation, no residual value.
- On-line consultations brought revenues of 300,000 € in 2016.
- Rented premises: 1,000 €/month from 1/01/2016. Salaries and Social Security of the permanent employees: 5,000 €/month. Electricity, phone, etc.: 500 €/month.
- Variable wages of the additional staff for the on-line consultations: 174,000 € in 2016.
- The partners contributed an initial capital of 100,000 €. A bank granted a loan of 100,000 € at 10% yearly interest for 5 years (1/01/2016 to 31/12/2020). Each 31 December the company returns 20,000 € and pays the interest (10%) on the loan value at the beginning of each year.
- Results are declared at 23:59 on 31 December (after all payments and receipts, including interest, wages, Social Security, etc.), and the corporation tax of 30% on the profits is paid immediately. Dividends of 40% of the profits after taxes are distributed, and the final balance is drawn at 24:00.

Required: (1) the income statement for 2016; (2) assuming the tax has been paid and the dividends distributed, the balance sheet at 24:00 on 31/12/2016; (3) the balance sheet at 24:00 on 31/12/2017, assuming no new investments in 2017 and that the retained earnings and the cash balance at the end of 2017 remained unchanged with respect to 2016; (4) the ROA and ROE ratios in 2017.

Setup: data, timeline and hypotheses

Item	Value
Capital	100,000 € (1/01/2016)
Loan	100,000 € at 10%; 20,000 € returned each 31/12; interest on the balance at the start of each year

Item	Value
Equipment	100,000 €, bought 1/01/2016, paid 1/04/2016, 5 years, no residual value
Revenues 2016	300,000 €
Fixed monthly costs	Rent 1,000; permanent staff 5,000; utilities 500 (€/month)
Variable wages 2016	174,000 €
Tax	30% of profits, paid at 23:59 on 31/12
Dividends	40% of profit after taxes, paid at 23:59 on 31/12

Hypotheses:

1. The statement says the results are declared "after having made all payments and receipts", so the 300,000 € of revenues are **collected** and all 2016 costs are **paid** within 2016.
2. The balance is drawn at 24:00, **after** paying tax, interest, the 20,000 € instalment and dividends: none of them remains as a liability.
3. For 2017 the statement gives no operating data; we assume the **same revenues and operating costs as in 2016**, which is the only reading that lets requirement 4 be computed. With retained earnings unchanged, the whole 2017 profit must have been paid out as dividends (shown in Phase 3).

Frames: classification of every operation (2016)

Operation	Income statement 2016	Cash 2016	Open at 24:00 on 31/12/2016
Capital	–	+100,000	Capital 100,000
Loan	–	+100,000	Loan 80,000 (after the instalment)
Equipment	Depreciation 20,000	−100,000 (on 1/04)	Equipment 100,000 − 20,000
Revenues	300,000	+300,000	–
Rent, staff, utilities, variable wages	252,000	−252,000	–
Interest	10,000	−10,000	–
Instalment	–	−20,000	(reduces the loan)
Tax	5,400	−5,400	–
Dividends	– (distribution)	−5,040	Retained earnings 7,560

The equipment is bought on credit on 1/01 and paid on 1/04: between those dates it is a debt to the supplier, but at 31/12 that debt no longer exists.

Derivation

Step 1. Depreciation and interest 2016

$$D = \frac{100{,}000 - 0}{5} \times \frac{12}{12} = 20{,}000\ €/\text{year}$$

$$I_{2016} = 100{,}000 \times 0.10 = 10{,}000\ € \quad \text{(loan value at the beginning of 2016)}$$

Step 2. Income statement 2016 (requirement 1)

Line	Computation	€
Revenues (on-line consultations)		300,000
– Rent	$12 \times 1{,}000$	12,000
– Permanent staff and Social Security	$12 \times 5{,}000$	60,000
– Electricity, phone, etc.	12×500	6,000
– Variable wages		174,000
– Depreciation	Step 1	20,000
= Income from operations	$300{,}000 - 272{,}000$	**28,000**
– Interest expense	Step 1	10,000
= Income before taxes	$28{,}000 - 10{,}000$	**18,000**
– Income taxes (30%)	$0.30 \times 18{,}000$	5,400
= Net income	$18{,}000 - 5{,}400$	**12,600**

Distribution [Slide 23]: dividends $= 0.40 \times 12{,}600 = 5{,}040\ €$; retained earnings (self-financing) $= 12{,}600 - 5{,}040 = 7{,}560\ €$.

Step 3. Cash at 24:00 on 31/12/2016

Inflows	€	Outflows	€
Capital	100,000	Equipment (1/04)	100,000
Loan	100,000	Rent + staff + utilities	78,000
Revenues collected	300,000	Variable wages	174,000
		Interest	10,000
		Loan instalment	20,000

Inflows	€	Outflows	€
		Tax	5,400
		Dividends	5,040
Total	**500,000**	**Total**	**392,440**

$$\text{Cash}_{2016} = 500{,}000 - 392{,}440 = 107{,}560\,€$$

Step 4. Balance sheet at 24:00 on 31/12/2016 (requirement 2)

The loan after the first instalment is $100{,}000 - 20{,}000 = 80{,}000$ €. The instalment due on 31/12/2017 (20,000 €) is within one year, so it is a **current liability**; the other 60,000 € are long-term [Slide 16].

Assets	€	Liabilities and equity	€
Equipment	100,000	Capital	100,000
– Accumulated depreciation	(20,000)	Retained earnings	7,560
Non-current assets	**80,000**	**Equity**	**107,560**
Cash	107,560	Long-term bank loan	60,000
Current assets	**107,560**	Short-term bank loan (due 31/12/2017)	20,000
Total assets	**187,560**	**Total liabilities and equity**	**187,560**

Step 5. Balance sheet at 24:00 on 31/12/2017 (requirement 3)

Item by item:

- **Equipment:** no new investment; two years of depreciation, $2 \times 20{,}000 = 40{,}000$; net value $100{,}000 - 40{,}000 = 60{,}000$ €.
- **Loan:** second instalment paid, $80{,}000 - 20{,}000 = 60{,}000$ €: 20,000 € current (due 31/12/2018) and 40,000 € long-term.
- **Cash and retained earnings:** unchanged by statement: 107,560 € and 7,560 €.
- **Capital:** unchanged, 100,000 €.

Assets	€	Liabilities and equity	€
Equipment	100,000	Capital	100,000
– Accumulated depreciation	(40,000)	Retained earnings	7,560
Non-current assets	**60,000**	**Equity**	**107,560**
Cash	107,560	Long-term bank loan	40,000
Current assets	**107,560**	Short-term bank loan (due 31/12/2018)	20,000
Total assets	**167,560**	**Total liabilities and equity**	**167,560**

Is the statement consistent? With the 2016 operations repeated (hypothesis 3), 2017 gives:

$$I_{2017} = 80{,}000 \times 0.10 = 8{,}000, \quad \text{IBT} = 28{,}000 - 8{,}000 = 20{,}000, \quad T = 0.30 \times 20{,}000 = 6{,}000, \quad \text{NI} = 14{,}000$$

Retained earnings unchanged means dividends $= 14{,}000$ € (a 100% pay-out). Cash then moves by $+300{,}000 - 252{,}000 - 8{,}000 - 20{,}000 - 6{,}000 - 14{,}000 = 0$, so the cash is indeed unchanged: the two conditions of the statement are compatible.

Step 6. ROA and ROE in 2017 (requirement 4)

> **DEFINITIONS NOT IN THE TOPIC 3 SLIDES**
> The Topic 3 deck in `sources/` does not define ROA and ROE, but the problem asks for them. The standard definitions used here are $\text{ROA} = \dfrac{\text{Income from operations}}{\text{Total assets}}$ (the return of the economic structure) and $\text{ROE} = \dfrac{\text{Net income}}{\text{Equity}}$ (the return to the owners). Check them against your class notes: if your lecturer defines ROA with net income instead, the value is $14{,}000/167{,}560 = 8.36\%$.

$$\text{ROA}_{2017} = \frac{28{,}000}{167{,}560} = 0.1671 = 16.71\%$$

$$\text{ROE}_{2017} = \frac{14{,}000}{107{,}560} = 0.1302 = 13.02\%$$

Interpretation and checks

- **Balances:** $80{,}000 + 107{,}560 = 187{,}560 = 107{,}560 + 60{,}000 + 20{,}000$ (2016) and $60{,}000 + 107{,}560 = 167{,}560 = 107{,}560 + 40{,}000 + 20{,}000$ (2017).
- **What changes between the two years:** both sides shrink by 20,000 €: the asset loses 20,000 € of value through depreciation, and the loan falls by the 20,000 € instalment, which in cash terms was financed by the cash generated (depreciation is an expense that is not a payment).
- **Leverage** [Slide 36]: $L/E = 80{,}000/107{,}560 = 0.74$ in 2016 and $60{,}000/107{,}560 = 0.56$ in 2017; the company becomes less indebted each year.
- **ROE < ROA here:** the return to the owners is lower than the return of the assets because interest and taxes are deducted from the numerator; the equity base being smaller than total assets is not enough to compensate.

- **Exam traps:** (1) the loan instalment is a payment, never an expense; (2) at 24:00 no tax or dividend liability remains; (3) the current part of the loan (next instalment) must be separated from the long-term part.

Check against the official key

`OFFICIAL SOLUTION, T3_ADDITIONAL_SOLUTIONS.PDF, PP. 12-14`

Requirements 1 and 2 coincide with the key in every figure: EBIT 28,000; interest 10,000; income before taxes 18,000; tax 5,400; net income 12,600; dividends 5,040; retained earnings 7,560; cash 107,560; equipment net 80,000; total 187,560; bank loan 20,000 current and 60,000 long term; equity 107,560.

The key does not solve requirement 3 (balance sheet at 31/12/2017) nor requirement 4 (ROA and ROE), so Steps 5 and 6 above are not covered by it.